



MotoDB Vehicle Sales Analysis

Cross-Platform KPI Validation & Interactive Dashboard Showcase

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Executive Summary & Data Pipeline

From raw auction records to a fully validated, analysis-ready dataset



Dataset Overview

550,296

Total Vehicle Sale Records

16

Features Analyzed per Record



Data Cleaning Pipeline

- **Row Repair**
Corrected 26 shifted rows misaligned during ingestion
- **Deduplication**
Removed all exact duplicate records
- **Outlier Imputation**
Price & odometer outliers imputed using Median values

✓ **100% Clean, Analysis-Ready Dataset**

Cross-Platform KPI Validation Matrix

Every KPI reconciled independently across Python, Excel, and Power BI

Metric	Validated Result	Python	Excel	Power BI
Total Revenue	\$7,484,689,599	✓ Match	✓ Match	✓ Match
Avg. Selling Price	\$13,601.21	✓ Match	✓ Match	✓ Match
Avg. MMR	\$13,762.34	✓ Match	✓ Match	✓ Match
Total Vehicles Sold	550,296	✓ Match	✓ Match	✓ Match
Avg. Odometer	68,480 miles	✓ Match	✓ Match	✓ Match

100% agreement across all five KPIs — the pipeline is validated end-to-end across every platform.

Key Analytical Insights

Univariate, bivariate & multivariate findings across the vehicle sales data



Market Volume

Sedans and SUVs dominate, together accounting for over 70% of total auction sales volume.



Transmission Pricing

Automatic vehicles consistently command higher average selling prices than manual options across top brands.



Price vs. MMR Gap

Actual selling prices tightly track MMR estimates, with only minor state-level variations.



Odometer Impact

Price depreciation accelerates sharply once mileage crosses the 60k-80k mile threshold.

Excel & Power BI Dashboards Showcase

Interactive tools built for stakeholders to explore the data themselves

Excel Pivot Dashboard



Pivot Tables & Charts

Sales by Region • Body Style • Price Bands

Dashboard preview — embed screenshot here

Power BI Interactive Dashboard



State-Level Map + Slicers

Filter by Make • Body Style • Sale State

Dashboard preview — embed screenshot here

Strategic Business Recommendations

Turning validated insights into concrete next steps



1. Inventory Focus

Increase acquisition of high-margin SUV and Sedan models with odometer readings under 60,000 miles.



2. Pricing Strategy

Utilize regional MMR variations to optimize reserve prices in top-performing sales states.



3. Digital Automation

Integrate automated data pipeline checks to prevent row-shifting and anomaly errors in future SQL uploads.

Questions & Discussion

Thank you for reviewing the MotoDB Vehicle Sales Analysis



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